Calcurion Maths Worksheets

A-Level Further Mathematics

Topic: Core Pure 1 – Argand Diagrams

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Plot the points representing $2 + 3i$, $-1 + 2i$ and $3 - i$ on an Argand diagram. **[3]**

2. Describe the locus $|z| = 3$. **[2]**

3. Describe the locus $|z - 2| = 4$. **[2]**

4. Describe the locus $\arg(z) = \frac{\pi}{4}$. **[2]**

5. Describe the locus $\arg(z - 1) = \frac{\pi}{3}$. **[3]**

6. Find the Cartesian equation of $|z - 1| = |z + 2|$. **[3]**

7. Find the Cartesian equation of $|z| = |z - 2i|$. **[3]**

8. Sketch the locus $|z - 1 - i| = 2$. **[3]**

9. Describe the region $|z| < 2$. **[2]**

10. Describe the region $\arg(z) \geq \frac{\pi}{6}$. **[2]**

11. Find the centre and radius of the circle $|z - 3 + 2i| = 5$. **[3]**

12. Find the distance between the points representing $1 + i$ and $4 - 2i$. **[2]**

13. Show that the points 1, i and $1+i$ form a right-angled triangle on the Argand diagram. **[4]**

14. Describe the locus $\mathrm{Re}(z) = 3$. **[2]**

15. Describe the locus $\mathrm{Im}(z) = -2$. **[2]**

Section B – Hard (Q16–25)

16. Find the Cartesian equation of $|z-1| = 2|z+i|$. **[5]**

17. Sketch the region defined by $|z-2| \leq 3$ and $\arg(z) \leq \frac{\pi}{4}$. **[5]**

18. Find the minimum value of $|z-3|$ given that $|z-1| = 2$. **[4]**

19. The point P represents z. Given $|z-2-i| = |z+1|$, find the Cartesian equation of the locus of P. **[4]**

20. Find the set of values of $\arg(z-1)$ if z lies on the line segment from $1+i$ to $1+3i$. **[4]**

21. Describe geometrically the transformation $z \mapsto iz$. **[3]**

22. Describe geometrically the transformation $z \mapsto \bar{z}$. **[2]**

23. Find the image of the line $\mathrm{Re}(z) = 1$ under the transformation $w = z + 2 - i$. **[4]**

24. The locus $|z-a| = k|z-b|$ is a circle (Apollonius). Find the centre and radius when $a = 1$, $b = -1$, $k = 2$. **[6]**

25. Sketch $|z-1| \geq |z+i|$ and shade the region. **[4]**

End of Worksheet • *Show all working clearly.*